

Metadata

Title

Island-Based Differences in Size Measurements of Female Adelie Penguins of the Palmer Archipelago

Description

This observational study seeks to investigate the size differences in female members of the Adelie Penguin (*Pygoscelis adeliae*) species within the islands of the Palmer Archipelago. Specifically, this study aims to analyze the difference in flipper length (mm), and body mass (g) between the female members of the Adelie species within the islands of the Palmer Archipelago: Biscoe, Dream and Torgerson.

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Study Information

Research questions

Research Question 1: Is there a difference in flipper length (mm) of female Adelie Penguins between the Biscoe, Dream and Torgerson islands of the Palmer Archipelago.

Research Question 2: Is there a difference in body mass (g) of female Adelie Penguins between the Biscoe, Dream and Torgerson islands of the Palmer Archipelago.

Hypotheses

Hypothesis 1 (non-directional): There is a mean difference in the bill length (mm) of female Adelie Penguins between the Biscoe, Dream and Torgerson islands of the Palmer Archipelago ($\mu_B \neq \mu_D \neq \mu_T$).

Hypothesis 2 (non-directional): There is a mean difference in the body mass (g) of female Adelie Penguins between the Biscoe, Dream and Torgerson islands of the Palmer Archipelago ($\mu_B \neq \mu_D \neq \mu_T$).

Data Description

Datasets used

Name and briefly describe the dataset(s), and if applicable, the subsets of the data you plan to use. Useful information to include here is the type of data (e.g., cross-sectional or longitudinal), the general content of the questions, and some details about the respondents. In the case of longitudinal data, information about the survey's waves is useful as well. Mention the most relevant information so that readers do not have to search for the information themselves.

The dataset being used in this study was acquired from *structural size measurements and isotopic signatures of foraging among adult male and female Adélie penguins (Pygoscelis adeliae) nesting along the Palmer Archipelago near Palmer Station* by the Palmer Station Antarctica LTER in 2020, and can be accessed [here](#). In the study, the data was collected seasonally (longitudinal) to examine ecological sexual dimorphism between Pygoscelis penguins as a result of foraging variability caused by changing levels of winter sea ice¹.

Data availability

All data and code are publicly available for this study and can be reached at [this](#) github repository.

Access date

The data for this project was accessed from the study above on 2026-03-11.

Data collection procedures

Detailed data collection methods can be viewed at the Palmer Station Antarctica LTER website or accessed [here](#).

—Still need to fill out the rest—

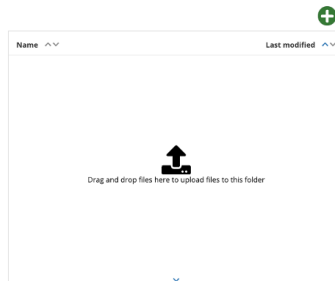
Variables

Manipulated variables

If you are going to use any manipulated variables from the study variables, identify them here. Describe the variables and the levels or treatment arms of each variable. Note that this is not applicable for observational studies and meta-analyses. If you are collapsing groups across variables this should be explicitly stated, including the relevant formula. If your further analysis is contingent on a manipulation check, describe your decisions rules here.

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Manipulated variables documentation



Measured variables*

Describe both outcome measures as well as predictors and covariates and label them accordingly. If you are using a scale or an index, state the construct the scale/index represents, which items the scale/index will consist of, and how these items will be aggregated. When the aggregation is based on exploratory factor analysis (EFA) or confirmatory factor analysis (CFA), also specify the relevant details (EFA: rotation, how the number of factors will be determined, how best fit will be selected, CFA: how loadings will be specified, how fit will be assessed, which residuals variance terms will be correlated). If you are using any categorical variables, state how you will code them in the statistical analyses.

Missing data

What do you know about missing data in the dataset (i.e., overall missingness rate, information about differential dropout)? How will you deal with incomplete or missing data? Provide descriptive information, if available, on the amount of missing data for each variable you will use in the statistical analyses. Based on this information, provide a new expected sample size.

Unit of analysis

Which units of analysis (respondents, cases, etc.) will be included or excluded in your study? Taking these inclusion and exclusion criteria into account, indicate the expected sample size of the data you'll be using for your statistical analyses. If you have a research question about a certain group you may need to exclude participants based on one or more characteristics. Be very specific when describing these characteristics so that readers will be able to redo your moves easily.

Statistical outliers

How will you define what a statistical outlier is in your data and what will you do when you encounter them? If you plan to remove outliers, provide a new expected sample size. If you expect to remove many outliers or if you are unsure about your outlier handling strategy, it is good practice to preregister analyses including and excluding outliers. Note that this will be the definitive expected sample size for your study and you will use this number to do any power analyses.

Sampling weights

Are there sampling weights available with this dataset? If so, are you using them or are you using your own sampling weights? Sampling weights can be useful in secondary data analysis because the sample may not be entirely representative of the population you are interested in.

Knowledge of Data

Prior Publication/Dissemination

List the publications, working papers, and conference presentations you have worked on that are based on the dataset you will use. For each work, list the variables you analyzed, but limit yourself to variables that are relevant to the proposed analysis. If the dataset is longitudinal, also state which wave of the dataset you analyzed. Specify the previous works for each co-author separately.

Prior knowledge*

Disclose any prior knowledge you may have about the dataset that is relevant for the proposed analysis. If you do not have any prior knowledge of it, please state so. Your prior knowledge could stem from

working with the data first-hand, from reading previously published research, or from codebooks. Provide prior knowledge for every author separately.

Analyses

Statistical models*

For each hypothesis, describe the statistical model you will use to test the hypothesis. Include the type of model (e.g., ANOVA, multiple regression, SEM) and the specification of the model. Specify any interactions and post-hoc analyses and remember that any test not included here must be labeled as an exploratory test in the final paper.

Effect size

If applicable, specify a predicted effect size or a minimum effect size of interest for all the effects tested in your statistical analyses.

Statistical power

Present the statistical power available to detect the predicted effect size or the smallest effect size of interest. Use the sample size after updating for missing data and outliers.

Inference criteria

What criteria will you use to make inferences? Describe the information you will use (e.g. specify the p-values, effect sizes, confidence intervals, Bayes factors, specific model fit indices), as well as cut-off criteria, where appropriate. Will you be using one- or two-tailed tests for each of your analyses? If you are comparing multiple conditions or testing multiple hypotheses, will you account for this, and if so, how?

Assumption Violation/Model Non-Convergence

What will you do should your data violate assumptions, your model not converge, or some other analytic problem arises?

Reliability and Robustness Testing

Provide a series of decisions or tests about evaluating the strength, reliability, or robustness of your finding. This may include within-study replication attempts, additional covariates, cross-validation, applying weights, selectively applying constraints in an SEM context (e.g., comparing model fit statistics), overfitting adjustment techniques used, or some other simulation/sampling/bootstrapping method.

Exploratory analysis

If you plan to explore your dataset to look for unexpected differences or relationships, describe those tests here. If reported, add them to the final paper under a heading that clearly differentiates this exploratory part of your study from the confirmatory part.

References

- (1) LTER, P. S. A.; Gorman, K. Structural Size Measurements and Isotopic Signatures of Foraging among Adult Male and Female Adélie Penguins (*Pygoscelis Adeliae*) Nesting along the Palmer Archipelago near Palmer Station, 2007-2009, 2020.
<https://doi.org/10.6073/PASTA/98B16D7D563F265CB52372C8CA99E60F>.